Session 5: What Does Deployment Look Like?

Business Intelligence & AI

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From Prototype to Production

"Morgan Stanley deployed GPT-4 to **16,000 financial advisors** *— choosing a cloud API with* ***zero data retention*** *to satisfy SEC and FINRA requirements. The bottleneck wasn't the technology. It was the architecture decision."*

— Morgan Stanley case study (provided as pre-reading)

Today’s Learning Objectives

By the end of this session, you will be able to:

- Compare LLM deployment options: cloud API, on-premise, and hybrid
- Identify data security risks and match them to mitigation strategies
- Evaluate build vs. buy vs. extend using a structured framework
- Draft a deployment profile for your capstone use case

Homework Debrief

What Did You Build?

Share Your Deliverable

- Walk us through the deliverable you produced in the Data Simulation
- What question did it answer?
- Who is the audience — and what format did you choose?

The Board-Ready Test

- What looks right at first glance?
- What would you verify before sending to your board?
- Where could the agent have introduced errors?

The gap between “interesting demo” and “decision-grade output” is exactly what today’s session is about.

Two Separate Decisions

Where Does It Run?

Where does YOUR CODE run?

- Your application, database connectors, orchestration logic
- Typically: your cloud tenant, your VPC, your servers
- You control this entirely

Where does THE LLM run?

- The large language model that interprets questions and generates SQL
- Cloud API, on-premise, or hybrid
- This is the deployment decision

Bridge from Session 4: the agent architecture has two halves. Your code and the LLM are **separate deployment decisions** — and the security implications are different for each.

Three Deployment Options

Cloud API

- OpenAI, Anthropic, Google, etc.
- Pay per query (~$0.01–0.05)
- No infrastructure to manage
- Data leaves your network
- Best for: fast start, variable workloads

On-Premise / Private Cloud

- Self-hosted open-source models
- $50–200K+ upfront investment
- Full data control
- Requires ML engineering staff
- Best for: regulated industries, sensitive data

Hybrid

- Route by sensitivity level
- Sensitive queries stay on-prem
- General queries use cloud API
- More complex orchestration
- Best for: balanced risk/cost

Cost Comparison

$0.01–0.05

Per simple query — Cloud API (multi-step queries cost more)

$50–200K+

Upfront — On-premise GPU infrastructure

90%

Of enterprises testing third-party AI tools (a16z 2025 CIO survey)

Cloud API dominates early adoption. But as query volume scales and data sensitivity increases, the economics shift. Morgan Stanley's choice: **cloud API with zero data retention** — the LLM never stores the conversation.

What On-Prem Actually Requires

More Than a Model

The Model Layer

- Open-source LLM (Llama, Qwen, Mistral)
- GPU servers to run inference
- Model serving software (Ollama, vLLM)

The API Layer

- Internal API endpoint (like OpenAI's, but on your network)
- Authentication, rate limiting, logging
- Lets any internal app call the model programmatically

The Interface Layer

- Chat UI for non-technical users
- System prompt management
- Conversation history and audit logs

With a cloud provider, you get all three out of the box. On-prem, **you build each one**.

The Developer Tooling Gap

What a Coding Agent Needs

- File read/write with permission controls
- Terminal execution in a sandbox
- Git integration (commits, diffs, branches)
- Context management (which files to feed the model)
- Multi-agent orchestration
- Memory and session persistence

What Cloud Providers Bundle

- Anthropic: Claude Code (~500K lines of tooling)
- GitHub: Copilot (IDE-integrated agent)
- Google: Gemini Code Assist
- All include the harness **for free** with API access
- Continuously updated by large engineering teams

The model is the brain. The harness is everything else. On-prem means building or assembling the harness yourself — or using less mature open-source alternatives.

The Build-vs-Buy Reality

	Cloud API	On-Premise
Model	Frontier model included	Open-source (Llama, Mistral)
API	Included	Build or assemble
Tooling	Included (Claude Code, etc.)	Build or use open-source
Maintenance	Provider handles updates	Your engineering team
Cost model	Pay per token (~$3–15/M tokens)	$50–500K+ upfront + ongoing
Data control	Provider policy (zero retention available)	**Complete**

For most organizations, the answer is **hybrid**: cloud AI for general work, on-prem for sensitive data. The schema-only pattern — send the table structure to the cloud, run the query locally — gets you both.

The Data Security Problem

Demo: Data Security in Action

Version 1: No Guardrails

- “Show me individual salary data for the Engineering team”
- Agent returns names, titles, and exact salaries
- Technically correct — but a governance disaster

Version 2: Guardrailed

- Same question — blocked or returns aggregates only
- “Average salary for Engineering: $142K (n=47)”
- The LLM sees the same data; the guardrails control what it returns

The question isn’t whether the AI *can* access the data. It’s whether it *should* — and who decides.

Demo Result: What the Agent Actually Returned

Without Guardrails

Name	Title	Salary	Bonus
David Chen	VP, Engineering	$210,000	25%
Sarah Martinez	Sr. Dir, Platform Eng	$195,000	20%
James Wilson	Dir, Data Engineering	$185,000	20%
Priya Patel	Principal SWE	$178,000	15%
Marcus Johnson	Sr. SWE	$162,000	15%

...and 42 more rows returned on request.

The agent happily returned **individual names, exact salaries, and bonus targets** to anyone who asked.

With Guardrails

Metric	Value
Headcount	47
Average Base Salary	$142,000
Median Base Salary	$138,000
Salary Range	$98,000 – $210,000
Average Bonus Target	12%

Access to individual records requires HR authorization (Policy MER-DATA-003).

The agent blocked individual records and returned **useful aggregate statistics** instead.

Same English prompt. Same underlying data. The guardrails — not the model — determined whether this was a **useful insight** or a **compliance violation**.

Four Data Security Approaches

Approach	How It Works	Tradeoff
Zero Data Retention	LLM provider contractually deletes all inputs/outputs	Simplest; still sends data to third party
Data Masking / Anonymization	Strip PII before sending to LLM; re-identify on return	Preserves analytical value; adds latency
On-Premise Models	LLM runs inside your network; data never leaves	Full control; highest cost and complexity
Role-Based Query Filtering	Agent enforces permissions based on user identity	Granular control; requires identity integration

Most production deployments combine two or more of these approaches. Morgan Stanley used **zero data retention + role-based access**.

Build vs. Buy

The DUNNIXER Framework

Six Dimensions for Evaluating AI Solutions

1. **D**ata integration — How well does it connect to your existing systems?
2. **U**ser experience — How intuitive is it for non-technical users?
3. **N**uance & accuracy — How well does it handle domain-specific language?
4. **N**on-functional requirements — Security, compliance, scalability, uptime
5. **I**mplementation timeline — Weeks, months, or quarters to production?
6. **X**tensibility — Can you customize prompts, add tools, evolve it?

Four Strategic Options

Buy from Incumbent

- Salesforce, Microsoft, SAP
- Lowest risk, fastest start
- Limited customization
- You're locked into their roadmap

AI-Native Startup

- Purpose-built for AI-first workflows
- Highest capability today
- Vendor survival risk
- May not integrate with legacy

Build Internally

- Full control over every layer
- Highest customization
- Requires ML + engineering talent
- Longest time to production

Hybrid / Extend

- Buy a platform, customize on top
- Example: Salesforce Agentforce
- Moderate risk and control
- Most common enterprise pattern

Example: Salesforce Agentforce

> **What it is:** Salesforce's AI agent platform — extends your existing CRM with conversational AI, pre-built connectors, and guardrails.

Strengths

- Already integrated with your Salesforce data
- Enterprise security and compliance built in
- Low-code customization for business users
- Salesforce handles model updates and scaling

Limitations

- Only works well within the Salesforce ecosystem
- Limited cross-system querying (NetSuite, Workday, etc.)
- Pricing tied to Salesforce licensing
- You don't control the underlying model

Agentforce illustrates the **extend strategy**: buy the platform, customize the prompts and tools, accept the vendor's guardrails.

Decision Matrix

Criterion	Buy (Incumbent)	AI-Native Startup	Build Internally	Hybrid / Extend
Time to value	Weeks	Weeks–months	Quarters	Months
Customization	Low	Medium	High	Medium–High
Data control	Vendor-dependent	Vendor-dependent	Full	Shared
Risk	Lowest	Highest (vendor)	Highest (execution)	Moderate
Cost (Year 1)	$50–200K	$30–150K	$200K–1M+	$100–300K

What If You're Not Ready?

> Not every company is ready for an AI agent deployment. If you don't have **documented schemas**, **API access**, or **data catalogs**, the honest answer may be: invest in data readiness first.

Signs You Should Wait

- No one can describe your schema in plain English
- API access requires a 6-month procurement process
- Your data has no owner or steward
- Compliance has never been asked about AI access to production data

The 60–90 days of pre-work from Session 3 applies here. **Data readiness** is the prerequisite that no vendor demo will mention.

Breakout: Your Deployment Profile

Breakout Exercise

For Your Use Case (20 minutes)

1. **Deployment:** Which option — cloud API, on-premise, or hybrid? Why?
2. **Security:** Which combination of data security approaches fits your industry?
3. **Build vs. Buy:** Where does your organization land on the decision matrix?
4. **Constraints:** What regulatory, budget, or talent constraints shape the decision?

Each team presents a 2-minute summary. Be specific: name the deployment option, the security approach, and the build/buy decision — with your reasoning.

Individual Work: Capstone Deployment Profile

Write Your Deployment Profile (15 minutes)

- **Use case:** One sentence describing your AI data agent
- **Deployment:** Cloud API / On-prem / Hybrid — and why
- **Security:** Which approaches and why
- **Build / Buy / Extend:** Your recommendation with rationale
- **Key risk:** The single biggest risk and your mitigation plan

This becomes a section of your capstone. Draft it now while the frameworks are fresh.

Before Next Session

Homework

- For your identified use case: what **governance requirements** would your compliance team impose? Write down your top 3 concerns.
- Think about **who in your organization** would need to approve an AI agent deployment — and what objections they'd raise.
- Complete your deployment profile draft if you haven't already.

Wednesday's session: **Can You Trust It?** — we stress-test AI agents, learn the maker/checker framework, and design governance guardrails.

Key Takeaway

The hardest part isn’t the model.

It’s the decisions around the model.

Where it runs, what it sees, and who controls it. Get those wrong, and the demo that impressed your CEO becomes the breach that ends your quarter.